

# Image-Native Language Models: Reading and Writing as Visual Generation

Imagized Language Model Project

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## Abstract

Most large language models represent language through symbolic text tokens and treat images of writing as an auxiliary input modality. This paper sketches a different research direction: an image-native language model whose inputs and outputs are visual canvases. The model learns from printed pages, handwritten forms, oracle-bone glyphs, bronze inscriptions, seal forms, and other historical scripts as images. It answers by generating readable images containing modern language and historically grounded glyph panels. We propose a 3090-friendly staged architecture based on visual autoencoding, masked visual language modeling, latent diffusion, and retrieval-augmented glyph evolution.

## 1 Motivation

Human readers do not see byte-pair tokens. They see pages, lines, strokes, spacing, damage, font variation, and script-specific visual forms. Current language models are extremely capable, but their central representation is normally a sequence of text tokens. Multimodal models accept images, and modern APIs also expose image-generation endpoints. For example, OpenAI documents APIs for processing images as input and generating images as output, and Google’s Gemini API documentation describes image generation from text and image prompts [4, 5, 6]. These systems are important baselines, but they do not by themselves define a language model whose primary language substrate is the visible page.

The Imagized Language Model (ILM) goal is different:

image input  $\rightarrow$  visual language state  $\rightarrow$  image output.

The output should itself be a readable answer image. For a query such as “explain the evolution of character zhong”, the system should produce a page containing modern explanatory writing plus oracle, bronze, seal, and modern forms.

## 2 Prior Art and Gap

OCR-free document models such as Donut and screenshot parsing models such as Pix2Struct show that document images can be understood without an explicit OCR pipeline. Token-free language models such as CANINE, ByT5, and MEGABYTE show that subword tokenization is not a hard requirement for language modeling. Discrete diffusion language models, including D3PM, Diffusion-LM, SEDD, and LLaDA-style masked diffusion, show that generation need not be left-to-right.

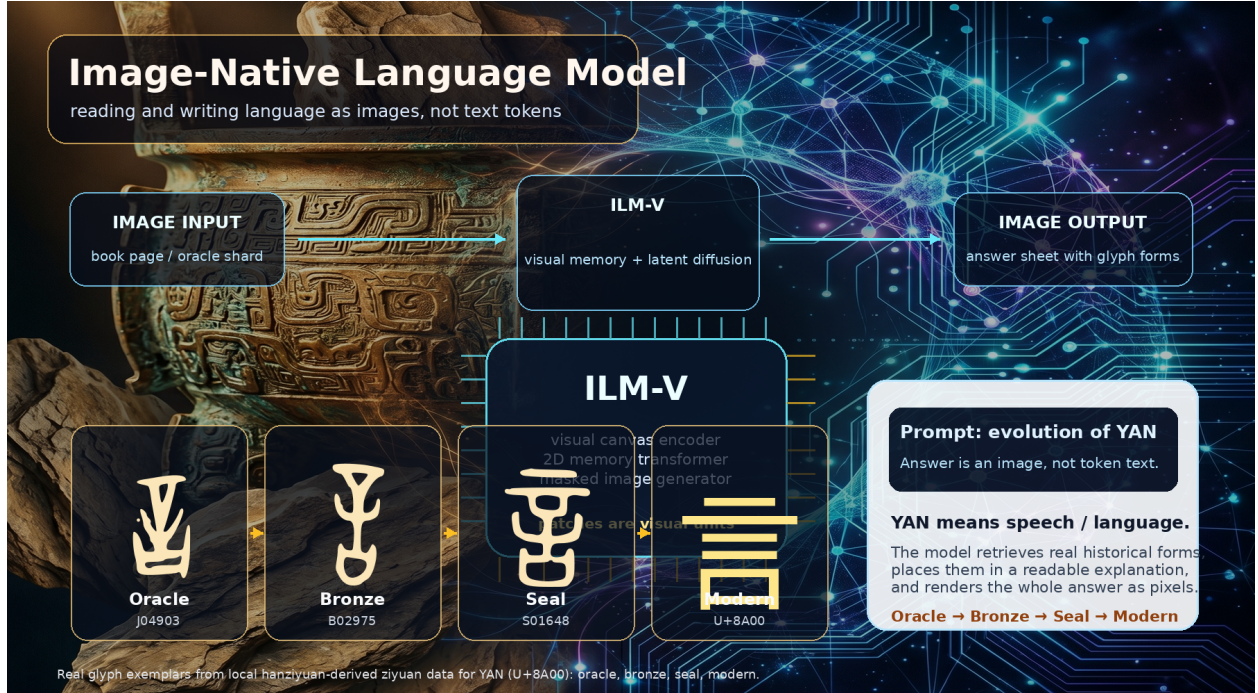


Figure 1: Image-native language modeling concept figure generated with an AgInTi background and overlaid with real local ziyuan glyph exemplars for YAN (U+8A00). The answer is a rendered image containing both modern explanation and historical forms.

Image generative models such as latent diffusion, MaskGIT, and DiT show that high-quality image generation can be performed in compressed latent spaces.

The missing combination is direct: train a model to read and write language as images, while preserving the visual structure of scripts and historical glyphs.

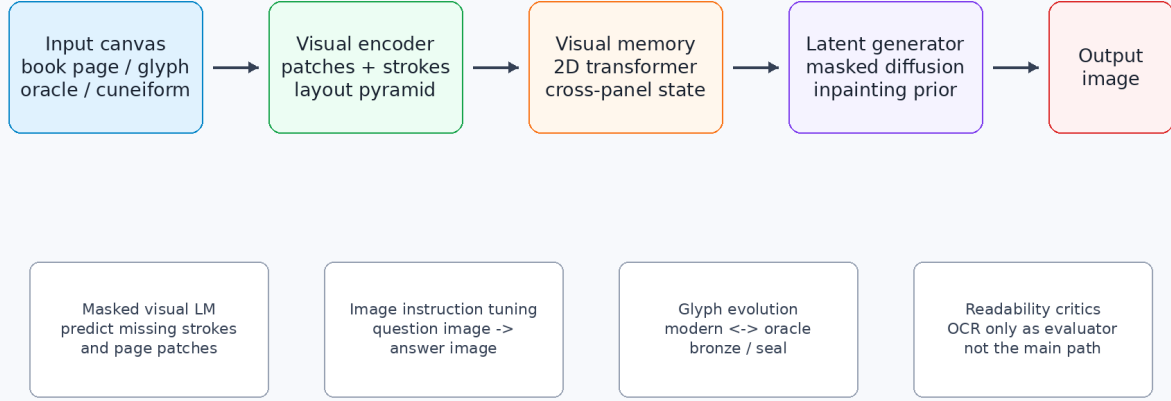
### 3 Architecture

Figure 2 gives the proposed system. The main components are:

1. **Visual encoder.** A patch/conv hybrid converts glyph tiles, line strips, and page crops into visual latent grids.
2. **Visual memory transformer.** A 2D Transformer maintains cross-panel state with relative position and sparse/window attention.
3. **Latent generator.** A masked diffusion or MaskGIT-like prior fills an output canvas conditioned on input canvas latents.
4. **Image decoder.** A VAE/VQ-VAE decoder converts latents back to readable output images.
5. **Auxiliary readers and critics.** OCR/readability heads evaluate the output, but should not become the primary hidden path.

The internal units are visual patches or visual codebook entries, not BPE, wordpiece, or Unicode language tokens.

## Image-Native Language Model (ILM-V)



**Key idea: patches/latents are visual units, not BPE or word tokens.**

Figure 2: ILM-V architecture. The model consumes input canvases, encodes them into visual latents, maintains a 2D visual memory, and denoises or fills output image latents. OCR-style readers are only auxiliary evaluators, not the main generation path.

## 4 Data

The project already has a strong historical Chinese glyph source outside the tracked repo. A local snapshot at `/home/lachlan/ProjectsLFS/incoder/data/historic` contains:

| Item          | Count  |
|---------------|--------|
| Characters    | 9,055  |
| Glyph records | 84,642 |
| Oracle        | 32,216 |
| Bronze        | 23,500 |
| Liushutong    | 21,923 |
| Seal          | 6,996  |

This is enough to train a first glyph autoencoder and a character-evolution generator. Modern book/page data can be added by rendering multilingual corpora into images and by ingesting scanned pages where licenses permit.

## 5 Training Objectives

The first ILM-V model should combine the following losses:

1. **Masked visual language modeling:** mask image patches and predict missing visual latents.
2. **Denoising reconstruction:** recover pages and glyphs from blur, noise, damage, dropout, and partial occlusion.

## Training ILM-V: Language as Visual Corpus

Every source is converted into page/glyph images. The model learns to predict and generate visual latents, not BPE or word tokens.

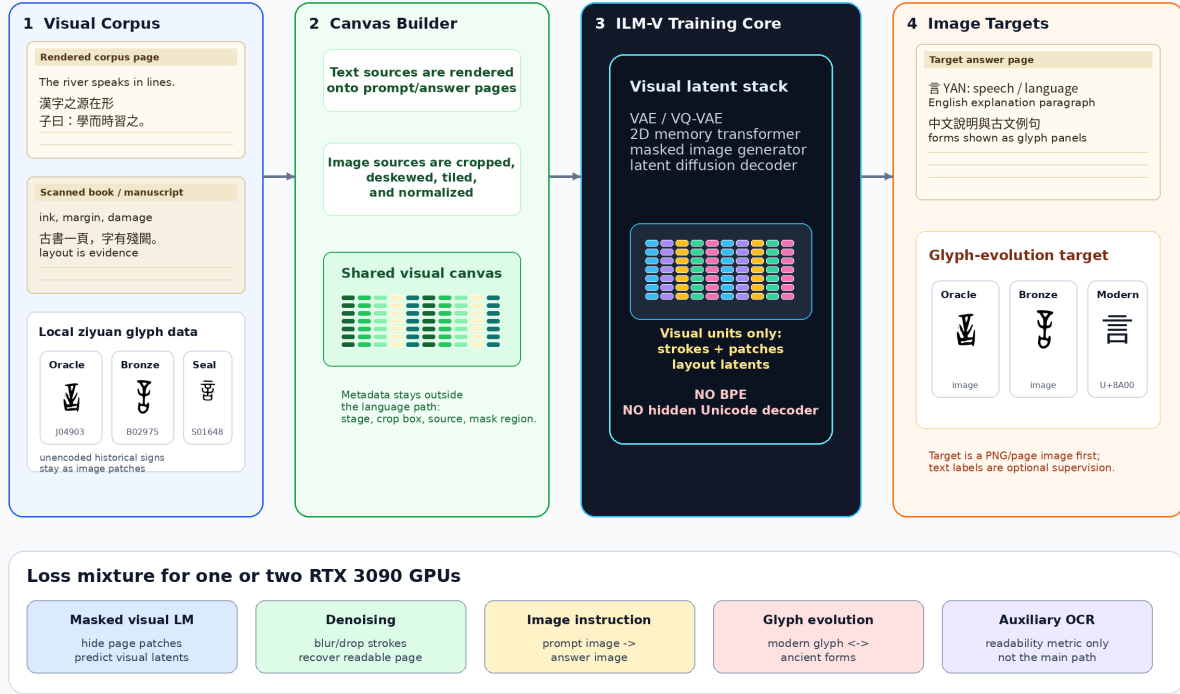


Figure 3: Training pipeline for ILM-V. Rendered text corpora, scanned pages, historical glyph files, and unencoded marks are all converted into visual canvases. The training target is visual latent prediction and image generation, not BPE-token prediction.

- Image-to-image instruction tuning:** render prompts and target answers as images and train conditional generation.
- Glyph evolution modeling:** map modern forms to historical stages and historical forms back to modern answer sheets.
- Contrastive visual semantics:** align different visual renderings of the same content without making OCR the bottleneck.

## 6 Image-First Training and Inference

Figure 3 makes the training contract explicit. Typed corpora are rendered into page images; scanned books and manuscripts are cropped and tiled; historical glyph files become glyph tiles; and marks that are missing from font tables remain valid image patches. Metadata such as crop boxes and stage labels may remain in manifests, but the model path should learn from visual latents. OCR is useful as a readability critic, not as the hidden language channel.

Figure 4 shows the user-facing path. A typed prompt is converted into a visual prompt canvas, and an uploaded image is normalized into the same visual space. The model returns a page-like image containing modern language and historical glyph regions. A later OCR/transcoder can attach

## Inference ILM-V: Chat-Like UI, Image-First Output

Typed text and uploaded images are both turned into visual prompt canvases. The model returns a rendered page image first.

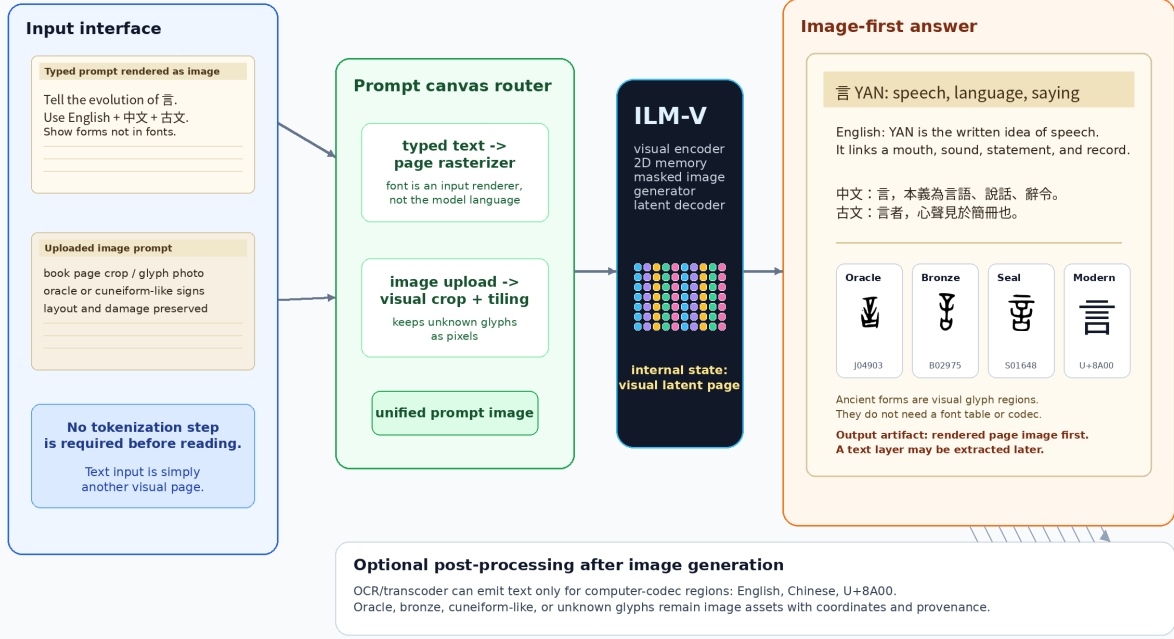


Figure 4: Inference pipeline for ILM-V. A chat-like interface may accept typed text or uploaded images, but typed text is first rasterized into a prompt image. The primary output is a rendered page image. OCR or transcoding is optional and only applies to output regions that exist in computer codecs.

a searchable text layer for English, Chinese, or Unicode-representable spans; ancient or unencoded glyphs remain image regions with coordinates and provenance.

## 7 3090-Friendly Curriculum

Figure 5 shows the practical schedule:

1. Build manifests for glyphs, rendered line images, and rendered instruction pairs.
2. Train a 30M–100M parameter glyph/page autoencoder.
3. Train a 100M–250M parameter conditional glyph evolution generator.
4. Train a 200M–600M parameter visual page language model on latent crops.
5. Add instruction tuning with image prompts and image answers.
6. Expand to additional scripts: cuneiform-like signs, hieroglyphic forms, Arabic, Japanese, Korean, and multilingual book pages.

Mixed precision, gradient checkpointing, latent crops, LoRA/adapters, and gradient accumulation make this path realistic on one or two 24GB GPUs.

## 3090-Friendly Training Curriculum

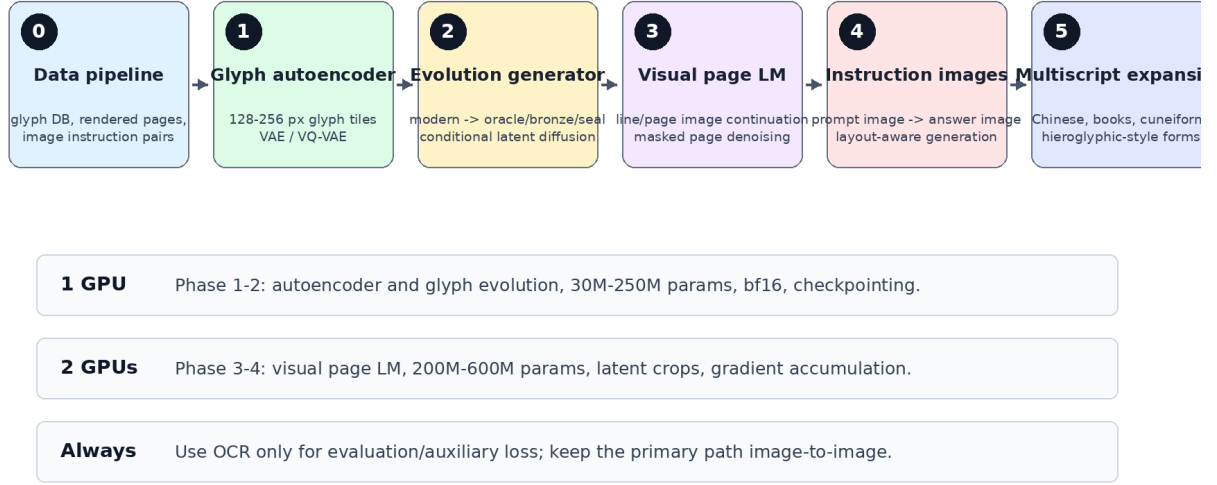


Figure 5: A staged curriculum that fits one or two RTX 3090 GPUs by starting with glyph tiles and latent generation before scaling to page-level image language modeling.

## 8 Target Demonstration

The first demo should answer a rendered visual prompt: “Explain the evolution of the Chinese character zhong.” The target answer image should contain:

- a title and concise modern explanation,
- a historical timeline with real glyph exemplars,
- stage labels and citations,
- modern-language text as rendered image.

This directly tests whether the model can combine language understanding and visual-script generation.

## 9 AgInTi Artifact Loop

The figure brief for AgInTi-style refinement is stored in the local AgInTi prompts subfolder. The current paper uses scripted PNG generation to keep the artifact reproducible.

## 10 Evaluation

The model should be evaluated as both a language model and a visual generator:

## Example Output: Evolution of Chinese Character Zhong (中)

A target answer is itself an image: explanation plus historical forms.



Figure uses local hanziyuan-derived glyph files when available; generation targets should retrieve/cite real exemplars.

Figure 6: Target output style for a glyph-evolution question. The answer is not merely text: it is a readable image with historical visual forms. The example uses local hanziyuan-derived glyph files when available.

- OCR readability of generated modern text images, used only as an external metric.
- Human readability and task-success ratings.
- Glyph-stage retrieval accuracy: generated glyphs should retrieve correct characters and stages.
- Historical plausibility: distance to nearest real glyph exemplars in stage-specific embedding space.
- Layout validity: no text/panel overlap, stable reading order, correct answer composition.
- Image-to-image QA accuracy on rendered prompts and rendered answers.

## 11 Conclusion

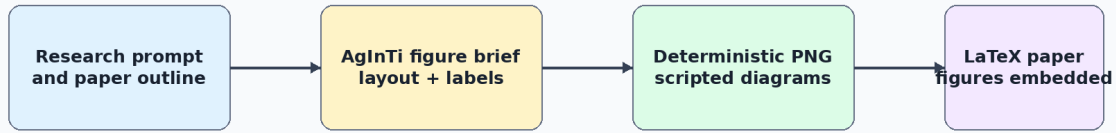
ILM-V is a feasible research direction because the pieces exist: OCR-free visual readers, token-free language models, visual latent autoencoders, masked image generators, and diffusion language models. The contribution is to make visible writing itself the modeling substrate. The local Chinese etymology dataset gives a concrete first domain where image output is essential rather than decorative.

## References

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- [3] OpenAI. GPT-4o API model documentation. <https://developers.openai.com/api/docs/models/gpt-4o>



## AgInTi-Assisted Research Artifact Loop



### Why scripted PNGs now?

They are reproducible, editable, and compile cleanly in the paper while leaving room for later AgInTi/image-model refinement

Figure 7: AgInTi-assisted artifact loop. The current figures are deterministic scripted PNGs so the paper is reproducible; the figure brief can later be sent to an interactive AgInTi/image agent for visual refinement.

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